High-fidelity libraries for enzyme optimization

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Abstract

Science abstract target: ∼125 words. Structure aligned to the three-application spine. Written last, on purpose — it should compress the actual results rather than lead them.

- **Hook.** De novo enzyme design produces backbones; optimization is the bottleneck that prevents most from reaching useful activity.
- **Approach.** Function-aligned RL fine-tuning of ProteinMPNN, combined with fragment-based combinatorial libraries and a closed-loop experimental optimization protocol.
- **Result (a) — heme oxidation.** Fine-tuned MPNN outperforms SSM and vanilla MPNN at matched experimental effort.
- **Result (b) — PETase.** Experimental data from an initial library trains a surrogate that drives a second library to substantially better designs (XXX-fold over starting; exceeds optimized natural enzymes on the PET-trimer probe).
- **Result (c) — protease.** A batch-consensus mutation-diversity reward unlocks rarer mutations and improves catalytic activity over an equivalent library without the diversity term.
- **Implication.** Broad, function-aligned sequence-space search plus an experimental closed loop is a general strategy for de novo enzymes.

[TODO: abstract — fill once results sections converge]

Contents

Document index

Every source file in `paper/`, what it holds, and how far along it is. **drafted** means real prose exists; **scaffold** means planning notes only; **needs data** means the section cannot be written until data lands.

File	Contents	Status
Build		
`main.tex`	Compile order; \input's everything	**drafted**
`preamble.tex`	Packages, palette, `\plan` / `\todo` macros, draft-mode switch	**drafted**
`Makefile`	`make pdf`; SVG→PDF via Inkscape	**drafted**
`refs.bib`	Bibliography; all entries machine-retrieved	**drafted**
Front matter		
`abstract.tex`	∼125-word abstract, written last	**scaffold**
`intro.tex`	Four-paragraph introduction arc	**scaffold**
`related_work.tex`	Status quo: backbone generation, filtering, RL for inverse folding, library design, the gap	**drafted**
Results — compile order is risk order		
`results/heme.tex`	Three-arm head-to-head; foundational claim	**needs data**
`results/petase.tex`	Closed-loop iteration; strongest data	**scaffold**
`results/protease.tex`	Diversity-reward ablation; highest risk	**needs data**
Methods		
`methods/overview.tex`	GRPO, fragment libraries, surrogate, diversity reward	**scaffold**
`methods/heme.tex`	Design config, three-arm protocol, assay	**scaffold**
`methods/petase.tex`	Two libraries, surrogate, assays	**scaffold**
`methods/protease.tex`	Design config, two-arm ablation, assay	**scaffold**
Back matter		
`conclusion.tex`	Four-paragraph discussion and outlook	**scaffold**

Two companion files live in `paper/` but are deliberately *not* part of this build:

- `outline_computational.md` — working brainstorm for a computational-first version of the paper: landscape analysis, the three candidate framings, seven proposed experiments (E1–E7), figure plan, and risk register. Section 2 is the LaTeX rendering of its landscape half.
- `figures/palette.py` — the plotting counterpart to the `cleo*` colors defined in `preamble.tex`.

Draft mode. This document currently compiles with `\draftmodetrue` in `preamble.tex`, which renders all planning notes in gray and all `\todo` markers in red. Setting `\draftmodefalse` hides every one of them, leaving only submission prose — useful for checking how much real text actually exists.

1 Introduction

Science research articles do not use `\section` labels in the main narrative. The headings here exist so that this draft has a usable table of contents; strip them at submission time. Paragraph-level scaffolding is kept visible so collaborators can see the intended arc.

P1 — The promise and the bottleneck

- RFdiffusion + MPNN make it possible to design enzyme backbones from scratch for diverse chemistries.
- First-pass designs are typically weak; the bottleneck is optimization, not initial design.

[TODO: p1 — frame de novo enzyme design + the optimization bottleneck]

P2 — Why existing optimization tools fall short

- Site-saturation mutagenesis (SSM): local; cannot escape the neighborhood of the starting design.
- Vanilla MPNN re-sampling: structurally informed but functionally blind; biased toward sequences MPNN's prior favors, not sequences that catalyze.
- The active-variant region for a de novo backbone often lies tens of mutations away — out of reach of either tool.
- Cross-reference Section 2 for the full treatment; this paragraph is the compressed version.

[TODO: p2 — limits of SSM and vanilla MPNN re-sampling]

P3 — What we contribute (three orthogonal capabilities)

- Function-aligned RL fine-tuning of ProteinMPNN (GRPO over reward pipelines) + fragment-based combinatorial libraries that can physically test combinatorially large candidate sets.
- **(i) Method foundation** — heme oxidation: fine-tuned MPNN beats SSM and vanilla MPNN at matched experimental effort.
- **(ii) Closed-loop iteration** — PETase: an initial experimental library trains a surrogate that becomes a reward signal for the next library, producing substantially improved designs.
- **(iii) Diversity-aware search** — protease: a batch-consensus mutation-diversity reward accesses rarer mutations and improves catalytic activity over an otherwise-matched library.

[TODO: p3 — three contributions, framed as separate claims]

P4 — Headline numbers + roadmap

- Heme: ⟨delta over SSM / vanilla MPNN at matched library size⟩.
- PETase: XXX-fold k_{cat} improvement on PET-trimer probe; best variant ∼30 mutations from starting; exceeds optimized natural enzymes on the same substrate.
- Protease: ⟨delta in activity for diversity-reward arm vs. baseline arm⟩.
- Together: a generalizable recipe for de novo enzyme optimization.

[TODO: p4 — headline results across the three applications]

2 Status quo: where the field stands and where the gap is

This section is written as real prose (not scaffold) because it is the part of the paper that does not depend on our own unfinished data. Every citation here was verified against Crossref, the arXiv API, or Europe PMC on 2026-08-04; the retrieval procedure is documented at the top of `refs.bib`. Quantitative claims attributed to other groups are taken verbatim from their abstracts.

2.1 Backbone generation is solved well enough that sequence design is now the bottleneck

Deep generative models now produce enzyme backbones for a wide range of chemistries. RFdiffusion established de novo design of protein structure and function by denoising diffusion over backbone coordinates [1]. RFdiffusion2 removed the requirement that catalytic geometry be specified at the residue level: it designs enzymes directly from functional-group geometries without specifying residue order or performing inverse rotamer generation, and generates scaffolds for all 41 active sites in a diverse in-silico benchmark, compared with 16 for previous methods [2]. Parallel efforts have produced designed serine hydrolases [3], metallohydrolases with designed metal sites [4], and enzymes built by catalytic motif scaffolding [5].

> The rhetorical work of this subsection is to concede generously. These are strong papers and several share our senior author. The point is not that backbone generation is inadequate — it is that its success relocates the bottleneck downstream, to sequence design and to what happens after the first 96 designs come back weakly active.

Two features of this literature matter for what follows. First, the sequence-design step is treated as a solved subroutine: backbones are passed to ProteinMPNN [6] or LigandMPNN [7] sampled at low temperature, and the resulting sequences are filtered against structure-prediction and geometric criteria. Second, the experimental readout is a small, one-shot test set: RFdiffusion2 reports active candidates after testing fewer than 96 sequences per catalytic mechanism [2]. Both features are appropriate for demonstrating that a backbone generator works. Neither addresses what to do when the resulting enzymes are, as they almost always are, orders of magnitude less active than useful.

2.2 Filtering is not search

The prevailing sequence-design protocol is generate-then-filter. Sequences are sampled from a functionally blind inverse-folding prior, then scored against structure-prediction confidence, backbone recapitulation, catalytic-geometry and preorganization criteria — AlphaFold3 [8], Boltz [9], and ensemble methods such as PLACER, whose introduction substantially improved experimental success rates in serine hydrolase design [3].

> This is the paper's central rhetorical move, so state it plainly and early, and make sure the phrasing survives review: a filter is a selection operator, not a search operator.

A filter cannot create a sequence that passes; it can only surface the sequences that a functionally blind prior happened to emit. This has two consequences that the field has largely left implicit. Sampling temperature is lowered to raise the pass rate, which means pass rate is purchased directly with diversity: ProteinMPNN's own report notes that sequence diversity can be increased considerably at higher temperature with only a small decrease in recovery [6], and the corresponding pass-rate cost is why production pipelines sit near $T = 0.1$. And because the survivors are drawn from a narrow neighborhood of the prior's mode, they are strongly correlated with one another — so a library's nominal size overstates the number of genuinely independent attempts it represents.

2.3 Reinforcement learning has reached inverse folding, but not libraries

Aligning generative models to objectives via reinforcement learning is now standard practice, and Group Relative Policy Optimization (GRPO) [10] in particular has proven to be a memory-efficient fit for the regime where rewards are computed by an external oracle rather than learned. This machinery has begun to reach protein sequence design. ProteinZero is an online RL framework for inverse folding that combines structural guidance from ESMFold with a self-derived $\Delta\Delta G$ predictor, adds an embedding-level diversity regularizer to mitigate mode collapse, and reports reducing design failure rates by 36–48% relative to ProteinMPNN, ESM-IF [11], and InstructPLM on the CATH-4.3 benchmark, with success rates above 90% across diverse folds [12]. Diversity-regularized direct preference optimization has been applied to peptide inverse folding with related motivation [13].

ProteinZero is the closest prior art to our mechanism and must be confronted in the first paragraph a reviewer reads on this topic, not buried. The differentiation is real but it is about the *objective*, not the optimizer: (i) general folds on CATH vs. enzyme active sites with ligands and catalytic geometry; (ii) sequence-level metrics — designability, recovery, stability — vs. a library-level objective; (iii) a diversity *regularizer* defended as anti-mode-collapse hygiene vs. diversity as a first-class design goal justified by screening economics; (iv) no physical library. If compute allows, running ProteinZero as a baseline arm is the strongest possible version of this argument. **[TODO: decide whether ProteinZero is runnable as a baseline]**

What this line of work does not address is the library. Its objectives and benchmarks are defined per sequence: is *this* sequence designable, stable, recovered. The quantity that determines an experimental outcome is a property of the set — how many distinct, genuinely different candidates clear the bar, and how far apart they are.

2.4 Library design assumes a natural enzyme

A mature literature does optimize libraries, but from the opposite starting point. MODIFY learns from natural protein sequences to infer evolutionarily plausible mutations and predict enzyme fitness, and explicitly co-optimizes predicted fitness and sequence diversity of starting libraries, prioritizing high-fitness variants while ensuring broad sequence coverage [14]. Active Learning-assisted Directed Evolution (ALDE) uses uncertainty quantification over an epistatic five-residue active site and improved the yield of a non-native cyclopropanation product from 12% to 93% in three rounds of wet-lab experimentation [15]. Machine-learning-guided cell-free expression platforms scale the assay side, mapping fitness across large variant sets [16].

MODIFY is the single most important citation for our diversity claim, because it proves the field already accepts that fitness and diversity should be co-optimized in a library. That is helpful — we are not arguing for diversity from scratch. What we are arguing is that every existing method for doing so is unavailable in the de novo regime.

Every one of these methods presupposes what a de novo enzyme lacks. MODIFY's fitness prior is evolutionary, learned from natural homologs; a de novo backbone has no homologs and no alignment. ALDE and active-learning methods require a measurable starting fitness to bootstrap from; de novo designs typically start at or near the detection limit. And both operate over a handful of positions, whereas the mutations that rescue a de novo enzyme are, in our hands, tens of substitutions from the starting design — a combinatorial volume that site-saturation and few-position libraries cannot reach.

2.5 The gap

Nobody has addressed library design for de novo enzymes. The regime is defined by three simultaneous absences: no evolutionary prior, so MSA-based fitness models do not apply; no starting activity, so active-learning loops have nothing to condition on; and a target region of sequence space too far away for local mutagenesis. The field's current answer — low-temperature inverse folding plus hard filters — is not a search procedure at all, and it responds to the difficulty of the regime by narrowing the search precisely when it should be widening it.

Close on the thesis so the reader carries it into the results. Draft formulation, to be tightened once F2 exists: function-aligned online RL converts in-silico filters from a rejection step into a design objective, producing de novo enzyme libraries that simultaneously pass the community's own fidelity criteria at higher rates and span far more sequence space — increasing the number of independent experimental bets per unit of screening effort.

[TODO: gap paragraph — final phrasing of the thesis sentence, once the headline figure is built]

3 Results: heme oxidation enzyme (method foundation)

Claim. Fine-tuned ProteinMPNN beats SSM and vanilla MPNN re-sampling at finding active variants under matched experimental effort. **Risk.** This is the foundational claim of the paper. If the comparison is not clean, the entire methods narrative is in trouble — this section needs the cleanest, most defensible head-to-head. Status: **needs data**.

Block 1 — Setup

- Backbone: ⟨de novo heme oxidation enzyme, RFdiffusion origin⟩.
- Starting design: ⟨activity baseline⟩.
- Three arms, matched library budget (N variants ordered/tested): (i) SSM around starting design; (ii) vanilla MPNN re-sampling on backbone; (iii) fine-tuned MPNN (this work) + fragment library.

[TODO: heme results — setup paragraph]

Block 2 — Headline figure

- Activity distribution across all three arms.
- Hit rate (fraction above starting design / above background).
- Best-variant comparison.

[TODO: heme results — main comparison figure + numbers]

Block 3 — Mutation distance / sequence-space analysis

- Where in sequence space the best variants from each arm live.
- Demonstrate the SSM and vanilla-MPNN arms are confined to a near neighborhood while fine-tuned MPNN reaches farther.

[TODO: heme results — sequence-space analysis]

Block 4 — Mechanism check (if available)

- Structural / spectroscopic / kinetic confirmation that the improvement is real catalysis, not artifact.

[TODO: heme results — mechanism check]

4 Results: PETase (closed-loop iteration)

Claim. Experimental data from an initial library can be used to train a surrogate that becomes a reward signal for a second library, producing substantially improved designs. **Risk.** The second library has to be meaningfully better than the first — not just predicted-better. Status: strongest data of the three applications.

Block 1 — Initial library

- De novo PETase backbone, RFdiffusion origin.
- Library 1: function-aligned RL sampling without surrogate.
- ∼6k designs tested; activity distribution on the PET-trimer fluorogenic probe.

[TODO: petase results — library 1 setup and outcomes]

Block 2 — Surrogate training

- MLP ensemble trained on (sequence, activity) pairs from library 1.
- Validation behavior; calibration of predicted vs. measured activity.

[TODO: petase results — surrogate training]

Block 3 — Library 2 with surrogate-as-reward

- Surrogate added as a reward step in the RL pipeline for library 2.
- Library 2 size, mutation-distance distribution from starting design.
- Activity distribution comparison: library 1 vs. library 2.

[TODO: petase results — library 2 design and outcome]

Block 4 — Best-variant deep dive

- XXX-fold k_{cat} improvement over starting design on the PET-trimer probe.
- Comparison to optimized natural PETases on the same substrate.
- PET-dimer activity confirmed.
- Honest framing of the bulk PET gap (substrate accessibility, not catalysis).

[TODO: petase results — best variant kinetics + PET-dimer + bulk-PET caveat]

5 Results: protease (diversity-reward ablation)

Claim. A batch-consensus mutation-diversity reward accesses rarer mutations and improves catalytic activity over an otherwise-matched library that lacks the diversity term. **Risk — highest in the paper.** It depends on wet-lab activity data for both arms. The load-bearing finding is "rare mutations *improve* activity," not merely "we sampled more of sequence space"; without that activity uplift this section reduces to a diversity-for-diversity's-sake observation. Status: **needs data**.

Block 1 — Setup

- De novo protease backbone.
- Two arms, matched library budget: (i) RL fine-tuned MPNN, standard reward set; (ii) same as (i) plus the batch-consensus mutation-diversity reward (`cleo.design.utils.mutation_diversity`).

[TODO: protease results — setup, both arms]

Block 2 — Sequence-space coverage

- Mutation-frequency spectra: arm (ii) should populate rarer mutations more heavily than arm (i).
- Position-level entropy / consensus-divergence statistics.

[TODO: protease results — sequence-space coverage figure]

Block 3 — Activity comparison

- Activity distribution: arm (i) vs. arm (ii).
- Best-variant comparison.
- Are the rare-mutation-bearing variants the ones with highest activity? This is the punchline — call it out explicitly.

[TODO: protease results — activity comparison; rare-mutation/activity link]

Block 4 — Mechanism for the diversity payoff

- Which rare mutations contributed; structural rationalization if possible.

[TODO: protease results — mechanistic rationalization]

6 Methods: shared pipeline

Shared methodology underlying all three applications. Per-application specifics live in the following subsections. Do not write this out in detail before the Results numbers exist — Methods text written against placeholder numbers gets rewritten.

Block 1 — RL fine-tuning of ProteinMPNN with GRPO

- Reference `cleo.design.train_policy` and `cleo.design.utils.grpo`.
- Reward pipeline composition (`UniversalReward`, ordered metric steps, normalized weighted aggregation).
- Cite GRPO [10] and ProteinMPNN [6].

[TODO: methods overview — GRPO + reward pipeline]

Block 2 — Fragment-based combinatorial libraries

- Sampling, fragment splitting, combinatorial recombination.
- DNA assembly via Golden Gate adapters.

[TODO: methods overview — fragment library construction]

Block 3 — Surrogate models for closed-loop optimization

- MLP ensemble with Gaussian NLL.
- Acquisition function (BatchUCB + diversity).
- Used as a reward step inside the RL pipeline.

[TODO: methods overview — surrogate ensemble + acquisition]

Block 4 — Batch-consensus mutation-diversity reward

- Per-position modal AA across the batch (ties multi-modal).
- Two metrics: `consensus_divergence` (count) and `fractional_divergence` ($1/k$ weighted, penalizes secondary-mode collapse).
- Reference `cleo.design.utils.mutation_diversity`.
- Contrast with the embedding-level diversity regularizer of ProteinZero [12] and diversity-regularized DPO [13]: ours acts in mutation space against a named reference sequence, theirs in embedding space.

[TODO: methods overview — diversity reward definition]

6.1 Methods: heme oxidation enzyme

Application-specific methods. Cross-reference Section 6 for shared pipeline details.

- **Block 1 — Design configuration.** Backbone, fixed residues, reward pipeline configuration.
- **Block 2 — Three-arm comparison protocol.** SSM library construction; vanilla MPNN re-sampling protocol (temperature, fixed residues); fine-tuned MPNN + fragment library; matched-budget rule — how N was chosen and held constant across arms.
- **Block 3 — Activity assay.**

[TODO: heme methods — design configuration] [TODO: heme methods — three-arm protocol] [TODO: heme methods — assay description]

6.2 Methods: PETase

- **Block 1 — Library 1 design configuration.** Backbone, fixed catalytic residues, library 1 reward pipeline (Boltz oracle [9] + PETase metrics + Hamming distance to reference).
- **Block 2 — Surrogate training.** Featurization, train/val split, ensemble size, calibration check.
- **Block 3 — Library 2 with surrogate-as-reward.** Reward pipeline composition for library 2.
- **Block 4 — Activity assays.** PET-trimer probe, PET-dimer, bulk PET attempt.

[TODO: petase methods — library 1 design configuration] [TODO: petase methods — surrogate] [TODO: petase methods — library 2 design configuration] [TODO: petase methods — assays]

6.3 Methods: protease

- **Block 1 — Design configuration.** Backbone, fixed residues.
- **Block 2 — Two-arm protocol.** Arm (i): standard reward set. Arm (ii): same plus the `mutation_diversity` reward. Record which variant was used — `consensus_divergence` or `fractional_divergence` — and the weight chosen. State the matched-budget rule.
- **Block 3 — Activity assay.**

[TODO: protease methods — design configuration] [TODO: protease methods — ablation protocol] [TODO: protease methods — assay description]

7 Discussion

Science research articles end with a short discussion / outlook paragraph rather than a separate Conclusion section. Keep this tight (~250–400 words). Write last, after the abstract.

P-A — Central scientific takeaway

- The active-variant region for a de novo enzyme is broader and farther from the starting design than SSM or vanilla MPNN can reach.
- Function-aligned RL sampling + combinatorial library construction + experimental closed loop together close the gap.

[TODO: conclusion p1 — central takeaway tying the three applications together]

P-B — What each application individually established

- Heme: fine-tuned MPNN dominates SSM and vanilla MPNN under matched experimental effort — the foundation.
- PETase: experimental data → surrogate reward → improved next library; a closed loop, not a single shot.
- Protease: explicit diversity pressure (batch-consensus divergence reward) unlocks rare mutations and improves activity over an otherwise-matched library.

[TODO: conclusion p2 — what each app proves about the platform]

P-C — Honest limitations

- PETase: probe and dimer activity demonstrated; bulk PET activity remains an open challenge attributed to substrate accessibility on solid plastic.
- Closed-loop iteration depends on having an assay-ready experimental pipeline; cold-start is still hard.
- Diversity reward improves access to rare mutations but does not substitute for a well-specified function reward.

[TODO: conclusion p3 — limitations, including bulk PET gap]

P-D — Outlook

- Generalization to additional chemistries and substrate classes, including bulk polymers.
- Tighter integration of design-time rewards with experimental surrogate signals across more rounds.

[TODO: conclusion p4 — outlook]

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